
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

21 Sep, 2026

Tutorial-5

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
-

Extreme Value Distributions and Extremes

Q1. Derivation of Exact and Asymptotic Extreme Value Distributions Let $X_1, X_2, \dots, X_n$ be a sequence of n independent and identically distributed (i.i.d.) random variables derived from an initial parent variate X with cumulative distribution function (CDF) $F_X(x)$ and probability density function (PDF) $f_X(x)$.

(a) Derive the exact CDF and corresponding PDF for:

(i) The sample maximum $Y_n = \max(X_1, X_2, \dots, X_n)$.

(ii) The sample minimum $Y_1 = \min(X_1, X_2, \dots, X_n)$ using $\Pr(Y_1 > y)$.

Note: the CCDF is also known as the survival function.

(b) **Type I (Gumbel Family):** Suppose the initial parent variate X follows an exponential distribution given by $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

(i) Show that as $n \rightarrow \infty$, the exact CDF of the sample maximum Y_n converges to the Gumbel Type I Maxima distribution:

$$F_{Y_n}(y) = \exp[-\exp(-\alpha_n(y - u_n))], \quad -\infty < y < \infty$$

and determine the location parameter u_n and scale parameter α_n in terms of n and λ .

(ii) Write down the general asymptotic CDF for the Type I Gumbel distribution for sample minima Y_1 .

(c) **Type II (Fréchet Maxima):** Consider a parent variate X with heavy power-law tail decay modeled by a Pareto distribution $F_X(x) = 1 - \left(\frac{x_0}{x}\right)^k$ for $x \geq x_0 > 0$ and shape parameter $k > 0$. Derive the exact CDF of the sample maximum $Y_n = \max(X_1, \dots, X_n)$, and show that as $n \rightarrow \infty$, it converges to the Fréchet Type II CDF:

$$G_{\max}(y) = \exp\left[-\left(\frac{v_n}{y}\right)^k\right], \quad y > 0$$

where $v_n = n^{1/k}x_0$ is the characteristic largest value. State the corresponding asymptotic expression for sample minima $G_{\min}(y)$.

(d) **Type III (Weibull Minima):** Consider a parent variate X bounded below by $\epsilon \geq 0$ whose CDF near the lower tail is modeled as $F_X(x) = \left(\frac{x - \epsilon}{w - \epsilon}\right)^k$ for $\epsilon \leq x \leq w$. Derive

the limiting asymptotic CDF for the sample minimum $Y_1 = \min(X_1, \dots, X_n)$ as $n \rightarrow \infty$, obtaining the Type III Standard Weibull distribution:

$$G_{\min}(y) = 1 - \exp \left[- \left(\frac{y - \epsilon}{w_1 - \epsilon} \right)^k \right], \quad y \geq \epsilon$$

where $w_1 = \epsilon + (w - \epsilon)n^{-1/k}$ is the characteristic smallest value. Explain why this distribution is uniquely suited for modeling material fracture strength (the “weakest-link” hypothesis).

Note on Fisher-Tippett-Gnedenko Theorem: By the extreme value limit theorem, normalized sample extremes $M_n = \max(X_i)$ or $L_n = \min(X_i)$ converge in distribution to only one of three asymptotic extreme value types (Gumbel, Fréchet, or Weibull) as $n \rightarrow \infty$, determined solely by the tail behavior of the parent CDF $F_X(x)$. ■

Q2. Extreme Sea Level Analysis and the Fukushima Paradox: Observed vs. Expected Maxima In coastal defense engineering, relying solely on short-term “observed sample maxima” often creates a dangerous illusion of safety because long-term “expected extreme maxima” calculated via Extreme Value Theory (EVT) can be substantially higher. During brief preliminary site observations at a coastal power facility, historical monitoring recorded a maximum sea surge level of only 5.7 m. However, the plant must be designed against rare, long-return-period tsunamis and storm surges.

Suppose the daily maximum sea swell height X follows an exponential parent distribution $F_X(x) = 1 - e^{-\lambda x}$ with parameter $\lambda = 0.85 \text{ m}^{-1}$ for $x \geq 0$. Consider $n = 365$ daily independent swell occurrences per year.

- Calculate the Gumbel Type I location parameter u_n (the mode or most probable annual maximum) and scale parameter α_n for the annual maximum sea surge level $Y = \max(X_1, X_2, \dots, X_{365})$. (*Hint: Use the expressions for u_n and α_n derived in Q1(b)(i).*)
- Determine the theoretical mean annual maximum surge level $\mathbb{E}[Y]$ and its standard deviation σ_Y . (*Hint: For a Type I Gumbel distribution, $\mathbb{E}[Y] = u_n + \frac{\gamma}{\alpha_n}$ where Euler’s constant $\gamma \approx 0.577216$, and $\sigma_Y = \frac{\pi}{\sqrt{6}\alpha_n}$.*)
- Derive the expression for the T -year return period surge level y_T . Calculate the 100-year (y_{100}) and 1000-year (y_{1000}) design sea surge levels. (*Hint: The annual exceedance probability for a T -year return period is $P(Y > y_T) = \frac{1}{T}$, which implies the annual non-exceedance probability is $F_Y(y_T) = 1 - \frac{1}{T}$.*)
- Contrast $y_{1000} \approx 15 \text{ m}$ with the short-term observed maximum of 5.7 m. Explain how underestimating extreme tails led to the breaching of coastal seawalls during events like the 2011 Fukushima Daiichi disaster.
- If a sea protection wall is built to 10.0 m, evaluate the probability that this sea wall will be overtopped by extreme sea surge at least once during the 50-year design life of the facility. (*Hint: Calculate the annual overtopping probability $p = 1 - F_Y(10.0)$ first, then determine the probability of overtopping at least once in 50 years.*) ■